

INTELLIGENT SYSTEM DEVELOPMENT
ASSIGNMENT 04

Comparing CNN and MLP Implementations: Scratch vs. Keras vs. PyTorch

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Three applications, one MLP and two CNNs, each built and trained three separate times, from scratch in NumPy, in TensorFlow/Keras, and in PyTorch, to compare abstraction level rather than to compare models against each other.

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1 Executive Summary

This report builds three intelligent systems, diabetes risk screening (tabular, binary classification), Fashion-MNIST clothing recognition (grayscale image, 10-class classification), and CIFAR-10 object recognition (RGB image, 10-class classification), each with a single architecture implemented three separate times: from scratch in NumPy with no deep learning framework, in TensorFlow/Keras, and in PyTorch. The point of the assignment is not to find the best of nine trained models. It is to test Lecture 04's central claim directly: that the same learning problem, expressed as the same architecture, should produce close to the same result regardless of which of the three abstraction levels builds it, and that what actually differs between the three is code visibility, developer convenience, and training speed, not the underlying mathematics.

Table 1 summarizes the three applications and which implementation won by test accuracy, the assignment's primary comparison metric.

Table 1: Executive summary across the three applications

Application	Data type	Task	Architecture	Winning implementation
Diabetes	tabular	binary classification	MLP (50 $\rightarrow$ 32 $\rightarrow$ 16 $\rightarrow$ 1)	PyTorch (0.782)
Fashion-MNIST	image, $28 \times 28 \times 1$	10-class classification	CNN (421,642 params)	Scratch (0.897)
CIFAR-10	image, $32 \times 32 \times 3$	10-class classification	CNN (545,098 params)	Keras (0.610)

Two results stand out ahead of the detailed sections below. First, the parameter count for every application is identical across all three implementations, confirmed by a direct assertion in each notebook (`assert keras_param_count == scratch_param_count`, and the same check against PyTorch), which is the most basic and most important sanity check this assignment makes: the three implementations really are the same architecture, not three architectures that happen to share a name. Second, the winning implementation changes from application to application, PyTorch on diabetes, scratch on Fashion-MNIST, Keras on CIFAR-10, and every winner's margin over the other two is small, at most a few accuracy points. That instability in the ranking, combined with the closeness of the scores, is itself evidence for Lecture 04's claim rather than against it: if one implementation reliably won every time, that would suggest a real difference in modeling power between scratch, Keras, and PyTorch, but a ranking that flips application to application is what three equally capable implementations of the same architecture should produce, differences settled by small, framework-level details, not by one implementation being fundamentally a better learner than the others. Training time tells a much less noisy story: Keras is fastest in every application, and the from-scratch NumPy implementation is dramatically slower than the other two specifically on the two convolutional networks, a real, measurable cost of visibility that the diabetes MLP, small enough that no implementation's overhead dominates, does not show.

2 Connection to Lecture 04: Same Architecture, Different Abstraction

This section is written once, at the framework level, so the three per-application sections that follow do not each have to re-derive the same argument. Every per-application section below instantiates the ideas explained here with its own concrete numbers and code, and references this section rather than repeating it.

2.1 The central question

Lecture 04 opens with a single question that frames the entire assignment: for three implementations of the same CNN, from-scratch NumPy, TensorFlow/Keras, and PyTorch, *what is the same, and what is different?* The lecture's own answer, stated directly, is that the conceptual architecture is unchanged across all three; what changes is only the level of abstraction. A Conv2d in scratch is an explicit function performing an `im2col` matrix multiply, written and read line by line. The same operation in Keras is one line, `Conv2D(...)`, with the multiply-and-accumulate hidden inside a compiled, optimized library routine. The same operation in PyTorch, `nn.Conv2d(...)`, sits in between: the layer itself is a one-line module like Keras's, but the training loop that calls it, and the backward pass that updates it, both stay explicit in the student's own code rather than being absorbed into a single `fit()` call.

2.2 Concept, not name, is what has to match

Lecture 04 states this plainly: different names do not imply different machine-learning concepts. Every implementation in this assignment expresses the same seven ideas, convolution (image apps only), activation, pooling (image apps only), flatten (image apps only), a dense/linear layer, a loss function, and an optimizer update, under a different name in each framework. Table 2 restates the lecture's own concept-to-API correspondence, trimmed to what this project actually uses; the full, per-application version of this table appears in each application's own comparison section below (slide 28's own deliverable), filled in with the exact code each implementation actually wrote rather than a generic placeholder.

Table 2: Concept-to-API correspondence used throughout this project

Concept	Purpose	Scratch	Keras / PyTorch
Convolution	local feature extraction	<code>conv2d_forward</code> (<code>im2col</code>)	<code>Conv2D</code> / <code>nn.Conv2d</code>
Activation	nonlinearity	<code>relu()</code>	"relu" / <code>nn.ReLU()</code>
Pooling	spatial reduction	<code>max_pool2d_forward</code>	<code>MaxPooling2D</code> / <code>nn.MaxPool2d</code>
Flatten	tensor $\rightarrow$ vector	<code>reshape</code>	<code>Flatten()</code> / <code>nn.Flatten()</code>
Dense/Linear	classification	<code>linear_forward</code> (mat-mul)	<code>Dense</code> / <code>nn.Linear</code>
Loss	measure error	manual formula	framework loss class
Optimizer	update parameters	<code>W -= lr * dW</code>	<code>optimizer="sgd"</code> / <code>torch.optim.SGD</code>

2.3 Where each implementation sits on the visibility spectrum

Scratch has maximum visibility: every multiply, every gradient, and every parameter update is code that was written by hand and can be read line by line, at the cost of the most implementation work and, as the results below show repeatedly, the slowest training time once real convolutions are involved. Keras has maximum convenience: the model definition is one `Sequential([...])` call, and `model.fit(...)` owns the entire training loop, the forward pass, the loss, the backward pass, and the parameter update, all invisible to the code that calls it. PyTorch sits deliberately in between: the layer definitions are as compact as Keras's inside an `nn.Module`, but the training loop (`zero_grad  $\rightarrow$  forward  $\rightarrow$  loss  $\rightarrow$  backward()  $\rightarrow$  step()`) stays explicit in the student's own code, with automatic differentiation (`autograd`) doing only the gradient computation, not the loop around it.

2.4 The fairness rule that governs every comparison below

Lecture 04 states the constraint under which any of these comparisons is meaningful: *same dataset, same split, same architecture, comparable hyperparameters*. Every application in this report follows that rule directly. For each application, the train/test split, the layer widths and depths, the batch size, the epoch count, and the learning rate are fixed once, before any of the three implementations is built, and reused unchanged across all three. The one place this project does not assume parity across frameworks is weight initialization: the scratch implementation uses explicit, seeded He initialization (`np.random.randn(...)* sqrt(2/fan_in)`, zero biases); Keras and PyTorch each apply their own default initializer, both also ReLU-appropriate and He/Kaiming-family, but not guaranteed to produce bit-identical starting weights to the scratch formula or to each other. That difference is treated as a legitimate, reportable source of divergence between implementations in the results below, not something to paper over or assume away.

3 Diabetes: Binary Classification (MLP)

3.1 Problem description

This application predicts whether a BRFSS (Behavioral Risk Factor Surveillance System) survey respondent has diabetes or pre-diabetes, from their self-reported health and demographic answers. The input is a flat feature vector with no meaningful 2-D spatial neighborhood between columns, BMI and marital status are not spatially related to each other the way neighboring pixels are, so this application uses a plain feedforward network (MLP) rather than a CNN, exactly the reasoning Lecture 04 gives for why Diabetes never gets a convolution: *there is no natural convolution operation* for tabular data of this kind.

3.2 Dataset

The dataset combines five full annual files from the Kaggle dataset

<https://www.kaggle.com/spandanjit2005/brfss-diabetes-indicator-dataset>

(DATASET_2016.csv through DATASET_2020.csv), intersected on the columns common to all five years (31 columns) and concatenated into a single table of **2,181,125 rows**. Two columns, `high_bp_00` and `high_cholesterol_00`, are present only in the 2017 and 2019 files, a structural difference in that year's questionnaire rather than a random missing-value problem, so both are excluded by the column intersection instead of being kept with three years of missing values.

The target is BRFSS's own diabetes field, originally three classes (non-diabetic / pre-diabetic / diabetic), collapsed to binary: 0 for non-diabetic, 1 for pre-diabetic or diabetic combined. The resulting target is meaningfully imbalanced, 1,841,657 non-diabetic rows against 339,468 diabetic-or-pre-diabetic rows, a 15.6% positive rate (Figure 1).

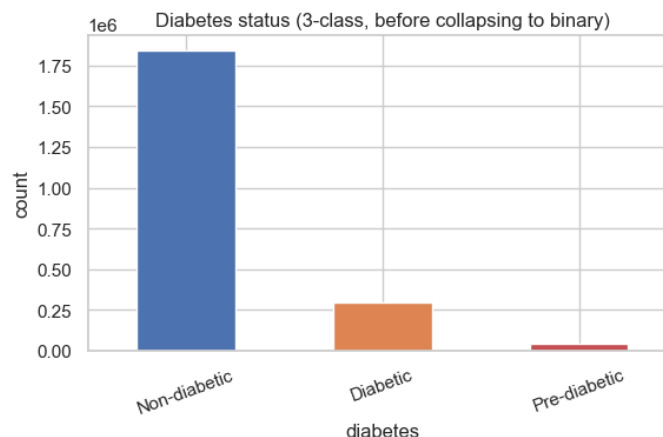


Figure 1: Diabetes status, three original classes, before collapsing to binary. Diabetic and pre-diabetic respondents are both a small minority next to non-diabetic respondents.

3.3 Data understanding and cleaning

BRFSS uses sentinel codes for "don't know" or "refused to answer" instead of leaving a cell blank, and those codes can sit far outside the range of a real answer. `weight_00` has a large spike exactly at 999 pounds, an essentially impossible real weight and a round, out-of-range value that would not happen by chance in genuine data, so every `weight_00 == 999` row (153,060 rows once counted precisely) is set to

missing rather than kept as a real measurement. The same reasoning applies to `bmi_00 > 100` (174,083 rows), a physically implausible body mass index. Both sentinel values are converted to missing before the train/test split, with the actual median imputation deferred until after the split so the imputed value is learned from training data only.

Missingness elsewhere in the dataset has a more mundane explanation: `avg_drinks_p_day_00` is missing for 52.5% of respondents and `poor_health_days_00` for 48.6%, both conditionally asked questions in the BRFSS questionnaire (average drinks per day only makes sense to ask someone who already said they drink), so the missingness reflects survey design, not data quality trouble. `income_00` is missing for 17.8% of respondents, a common pattern for income questions in surveys.

Several columns turned out to be string duplicates of a clean numeric ordinal column already present elsewhere in the table (`age_group_00` next to `age_00`, `education_level_00` next to `education_00`, `income_level_00` next to `income_00`, `general_health_00` next to `gen_health_00`, `last_checkup_00` next to `l_checkup_00`); keeping both would either double-count the same signal or require re-encoding the string version from scratch for no benefit, so the five string duplicates are dropped and the numeric ordinal versions are kept.

Exploratory analysis confirmed clinical relationships the model is expected to pick up on. Median BMI climbs from non-diabetic to pre-diabetic to diabetic respondents (Figure 2), matching the established link between excess body weight and type 2 diabetes risk. Self-reported general health moves almost monotonically with diabetes status (Figure 3): respondents rating their health "excellent" report diabetes or pre-diabetes at a much lower rate than those rating it "poor," which is also why this ordinal field is kept as a numeric code rather than one-hot encoded, since the ordering itself carries signal. The diabetic-or-pre-diabetic rate by survey year (Figure 4) drifts mildly across the five years, from 15.20% (2020) to 15.97% (2018), too small a gap on its own to argue strongly for keeping year as a feature, but kept anyway since it is one of the few signals left carrying any residual survey-wave information once `high_bp_00` and `high_cholesterol_00` had to be dropped for the schema mismatch described in Section 3.

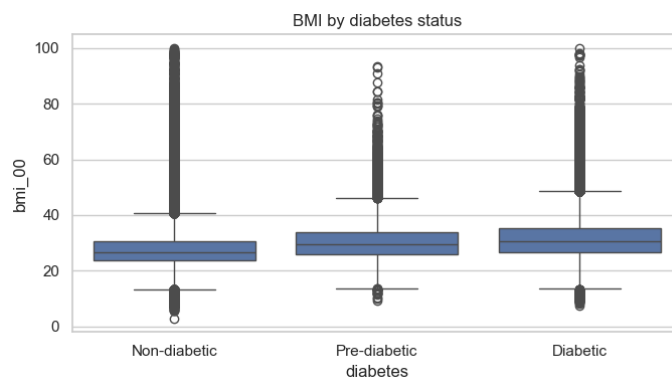


Figure 2: BMI by diabetes status. Median BMI rises step by step from non-diabetic to pre-diabetic to diabetic.

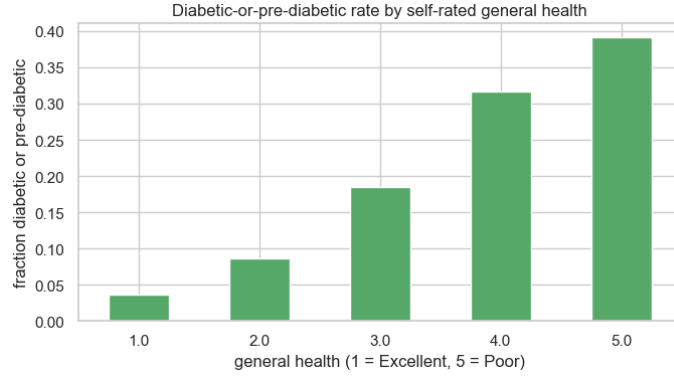


Figure 3: Diabetic-or-pre-diabetic rate by self-rated general health (1 = Excellent through 5 = Poor). The rate rises steadily from Excellent to Poor.

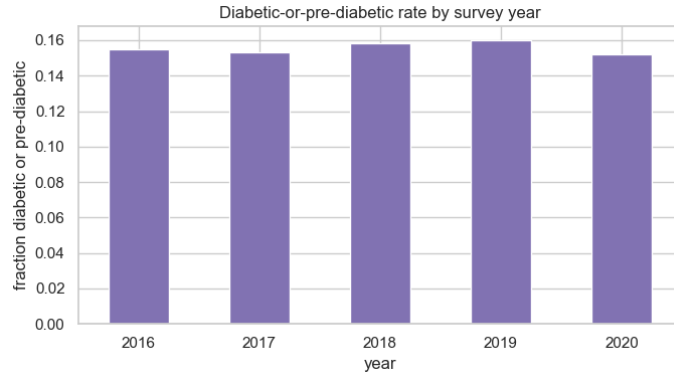


Figure 4: Diabetic-or-pre-diabetic rate by survey year, 2016-2020. The rate drifts mildly rather than sitting flat, the reason year is kept as a feature rather than dropped.

3.4 Representation

The remaining columns fall into three groups, each needing a different treatment before a neural network can use them. Eight genuinely two-valued columns (sex, exercise, smoking history, alcohol use, personal history of stroke, heart attack, or coronary disease, cost barrier to care) map directly onto 0/1 with no information lost. Four genuinely unordered categorical columns (race, marital status, employment status, personal-doctor access) are one-hot encoded, since turning "Married" into a single ordinal number would imply an ordering between marital statuses that does not exist; this produces 29 one-hot columns. The remaining 13 numeric columns, the ordinal survey codes (general health, education level) alongside genuinely continuous measurements (BMI, weight, height) and year itself, all go into one `StandardScaler` group fit on the training split only.

year is deliberately included in that scaled numeric group rather than left as the raw calendar integer. Left unscaled, year (2016 through 2020) would sit at a scale several times larger than every other feature, dominating the first layer's weighted sum before training has any real chance to learn something better than always predicting the majority class. Standardizing it alongside the other numeric columns removes that risk without discarding whatever residual survey-wave signal it carries.

This produces a feature matrix of width **50** (8 binary + 29 one-hot + 13 standardized numeric). The 80/20

stratified split (`RANDOM_SEED = 42`) gives 1,744,900 training rows and 436,225 test rows, with the 15.56% positive rate preserved in both splits.

Because the target is imbalanced, a plain fit would be trained to minimize an average loss dominated almost entirely by the majority class. Balanced sample weights, $w = n_{\text{total}} / (2 \times n_{\text{class}})$, give $w_{\text{neg}} = 0.5922$ and $w_{\text{pos}} = 3.2126$, so a mistake on a diabetic row costs roughly 5.4 times as much in the training objective as a mistake on a non-diabetic row. All three implementations apply this same balanced weighting, though, as Section 3.9 shows, not through identical mechanics underneath a shared name.

3.5 Shared architecture

All three implementations build the same network, computed at runtime from the confirmed input width $d = 50$ rather than hardcoded:

$50 \rightarrow \text{Linear}(32) \rightarrow \text{ReLU} \rightarrow \text{Linear}(16) \rightarrow \text{ReLU} \rightarrow \text{Linear}(1) \rightarrow \text{Sigmoid}$

with shape trace, for a batch of size B :

$(B, 50) \rightarrow (B, 32) \rightarrow (B, 32) \rightarrow (B, 16) \rightarrow (B, 16) \rightarrow (B, 1) \rightarrow (B, 1)$

Batch size 2048, 25 epochs, learning rate 0.1, and plain SGD (no momentum, no Adam) are fixed once here and reused unchanged by all three implementations, so the optimizer’s own update rule is identical across all three rather than needing a from-scratch Adam implementation to keep the comparison fair.

3.6 Three implementations

3.6.1 From scratch, NumPy only

Every component is an explicit Python function: `relu()`, a manual `binary_cross_entropy(y_true, y_pred, weights=None)` that accepts the per-row balanced weights, He-initialized weight matrices (`init_params`), an explicit forward that chains two `Linear` + `ReLU` blocks into a final `Linear` + `Sigmoid`, and a backward pass deriving every gradient by the chain rule, with the balanced sample weights multiplied into the output-layer error signal $dZ_3 = (A_3 - y) \odot w$ so imbalance correction propagates through every downstream gradient exactly as intended. A numerical gradient check confirms the hand-derived backward pass against a finite-difference approximation before any full training run. Mini-batch gradient descent (batch size 2048) trains for 25 epochs, one forward pass, one backward pass, one parameter update per batch.

3.6.2 TensorFlow/Keras

The identical architecture becomes a five-line `Sequential` definition (`Dense(32, "relu"), Dense(16, "relu"), Dense(1, "sigmoid")`), compiled with `SGD(learning_rate=0.1)` and `"binary_crossentropy"` loss, and trained with one `model.fit(..., class_weight={0: 0.5922, 1: 3.2126})` call. `class_weight` is Keras’s own mechanism for the same balanced weighting used by hand in scratch: it multiplies each class’s contribution to the loss by the same `weight_neg/weight_pos` values, so this implementation trains on the same effectively rebalanced problem as scratch, not a plain unweighted version of it. `model.summary()` reports 2,177 parameters, matching the scratch parameter count exactly and confirmed by a direct assertion in the notebook.

3.6.3 PyTorch

The same architecture becomes an `nn.Module` (DiabetesMLP) whose forward chains three `nn.Linear` layers with `nn.ReLU` in between, returning raw logits with no sigmoid applied inside the module. `nn.BCEWithLogitsLoss(pos_weight=...)` applies the sigmoid and the cross-entropy loss together in one numerically stable step, rather than computing them separately the way scratch and Keras do, matching the lecture's own guidance to let a framework's fused loss function handle the activation internally. `pos_weight` is set to $w_{\text{pos}}/w_{\text{neg}} \approx 5.42$, PyTorch's own mechanism for balanced weighting, applied to the training loop as an explicit `zero_grad` $\rightarrow$ forward $\rightarrow$ `loss.backward()` $\rightarrow$ `optimizer.step()` cycle rather than a single `fit()` call. `torch.optim.SGD` with the same learning rate closes the loop. The resulting parameter count, 2,177, matches both other implementations exactly.

3.7 Component-by-implementation table

Table 3: Diabetes: component-by-implementation table

Component	Scratch	Keras	PyTorch
Data loading	<code>pandas.read_csv</code> <code>pd.concat</code>	$\times 5$, same	same
Preprocessing	manual mean/std standardization	StandardScaler-equivalent (manual, shared)	same as scratch
Model definition	explicit functions + a params dict	<code>Sequential(...)</code>	<code>nn.Module</code> subclass
Activation	<code>relu()</code> , <code>sigmoid()</code>	<code>activation="relu"/"sigmoid"</code>	<code>nn.ReLU()</code> , fused in <code>BCEWithLogitsLoss</code>
Dense/Linear layer	<code>linear()</code> (manual matmul)	<code>Dense(...)</code>	<code>nn.Linear(...)</code>
Loss	manual <code>binary_cross_entropy</code>	<code>"binary_crossentropy"</code>	<code>nn.BCEWithLogitsLoss()</code>
Gradient	manual backprop	autodiff (implicit in fit)	<code>loss.backward()</code> (autograd)
Optimizer	manual $W \leftarrow W - lr * dW$	<code>optimizer="sgd"</code>	<code>torch.optim.SGD(...)</code>
Training loop	explicit for epoch / for batch	<code>model.fit(...)</code>	explicit for epoch / for batch
Evaluation	manual metric functions	<code>model.evaluate</code> -equivalent (manual)	manual (<code>model.eval()</code> , no-grad)
Prediction	<code>forward(X_test, params)</code>	<code>model.predict(...)</code>	<code>model(x_test)</code> under <code>torch.no_grad()</code>

3.8 Training curves and evaluation

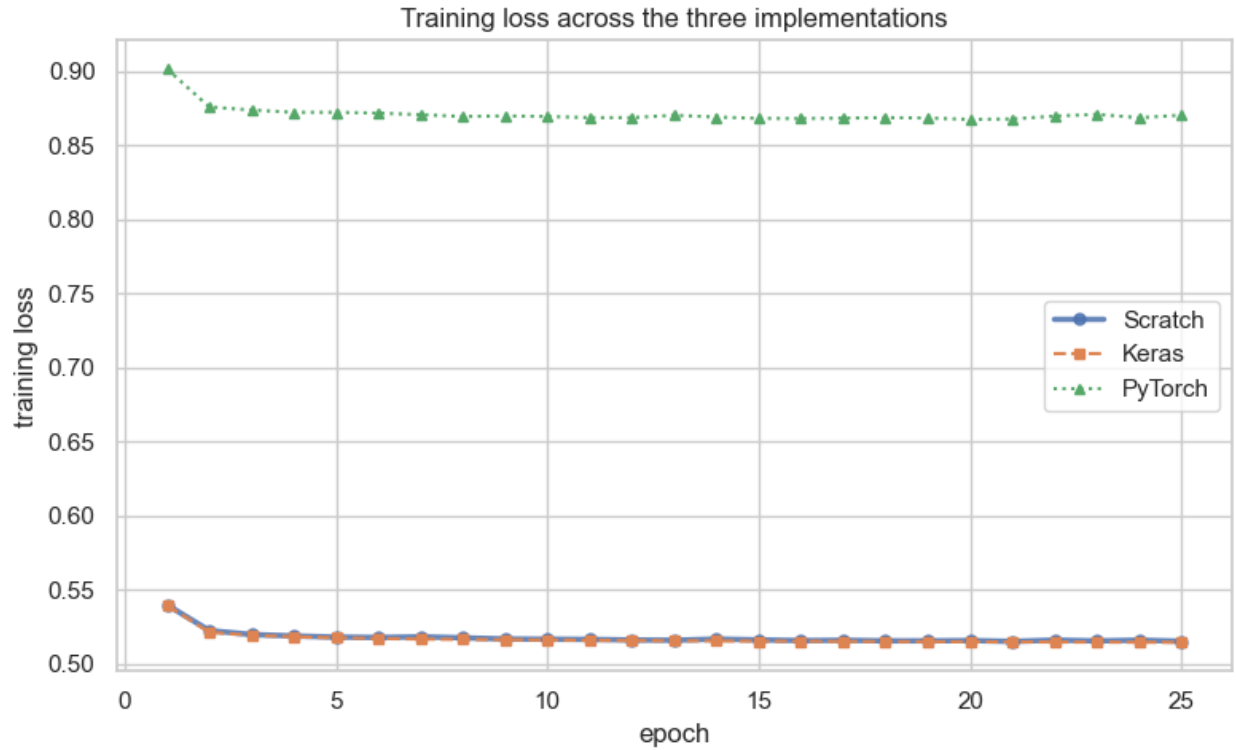


Figure 5: Training loss across the three implementations, one line per implementation. Scratch and Keras track each other closely; PyTorch’s curve sits on a visibly different numeric scale for reasons explained in Section 3.9.

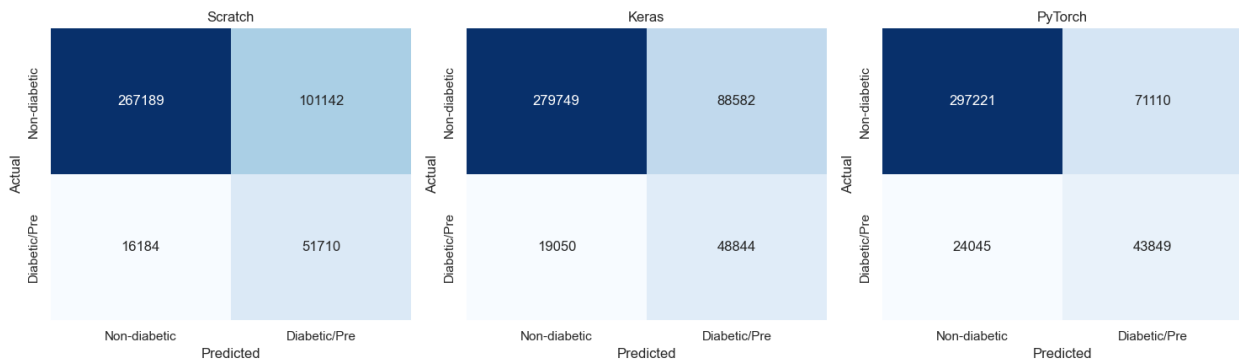


Figure 6: Confusion matrices for the three implementations on the diabetes test set (436,225 rows).

Table 4: Diabetes: quantitative comparison, test set

Implementation	Params	Train time (s)	Final loss	Accuracy	Precision	Recall
Scratch	2,177	69.4	0.5155	0.7310	0.3383	0.7616
Keras	2,177	24.7	0.5148	0.7533	0.3554	0.7194
PyTorch	2,177	311.0	0.8706	0.7819	0.3814	0.6458

F1 for the same three implementations, in the same order: 0.4685 (scratch), 0.4758 (Keras), 0.4796 (PyTorch).

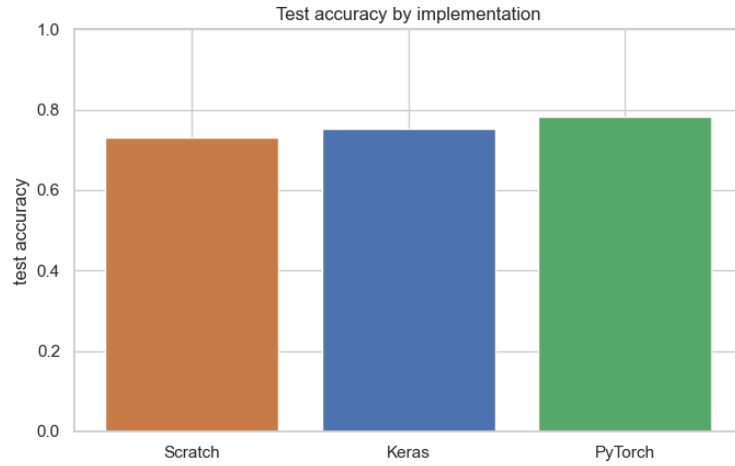


Figure 7: Test accuracy by implementation.

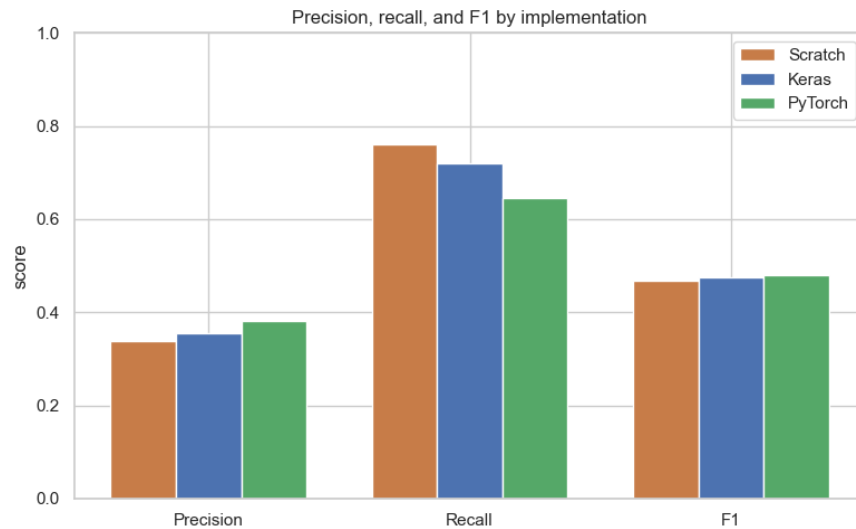


Figure 8: Precision, recall, and F1 by implementation. All three land in the same 0.34-0.38 precision, 0.65-0.76 recall range, the expected footprint of balanced weighting on an imbalanced target: more of the rare positive class gets caught, at the cost of more false positives.

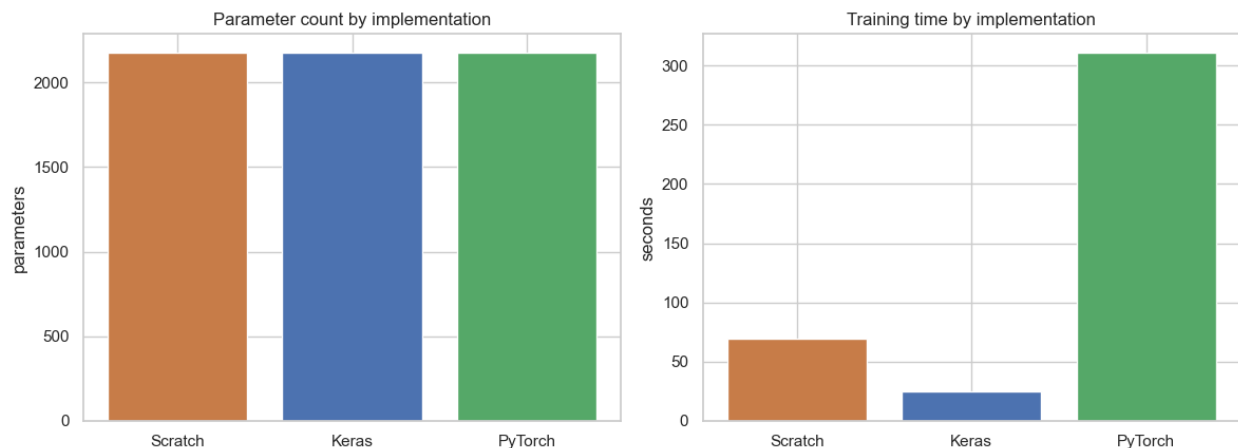


Figure 9: Parameter count (identical across all three, confirming the same architecture) and training time by implementation.

3.9 Comparison discussion

By accuracy and F1, PyTorch comes out slightly ahead (0.782, 0.480), Keras is close behind (0.753, 0.476), and scratch trails by a similar small margin (0.731, 0.469). All three land within about five points of each other on both metrics, the expected outcome for three implementations of the same architecture trained on the same data with the same hyperparameters: the gap is small enough to plausibly come from framework-level details (default numerical precision, exactly how each framework shuffles batches, small differences in how the balanced weighting is applied) rather than from one implementation being a meaningfully better model than the others.

One number in Table 4 is genuinely not comparable across implementations, and it is worth explaining rather than glossing over: the final training loss. Scratch and Keras both land around 0.51, while PyTorch sits at 0.87, a difference far bigger than the accuracy gap would suggest. The gap comes entirely from how each implementation applies balanced class weighting, not from PyTorch training worse. Scratch and Keras both scale *every* row’s loss, multiplying the negative class’s rows by $w_{\text{neg}} = 0.5922$ and the positive class’s rows by $w_{\text{pos}} = 3.2126$, which keeps the loss on roughly the same numeric scale an unweighted loss would have had. PyTorch’s `BCEWithLogitsLoss(pos_weight=...)` only scales the positive class’s term, by the ratio $w_{\text{pos}}/w_{\text{neg}} \approx 5.42$, leaving the negative class’s term unscaled. That is a legitimate, common way to apply class weighting in PyTorch, but it produces a loss value on a different scale, not a directly comparable number. This is exactly the kind of implementation-level detail this assignment is built to surface: three frameworks offering “the same” class-weighting feature under names that suggest equivalence (`class_weight`, a `weights` argument, `pos_weight`), while actually parameterizing it differently underneath.

Training time tells a different story than accuracy does. Keras was fastest (24.7s), scratch was next (69.4s), and PyTorch was noticeably slower (311.0s), close to four and a half times scratch’s time for a network with only 2,177 parameters. This lines up with how each implementation spends its time: Keras’s `fit()` runs its training loop in a compiled graph with very little Python overhead per batch; scratch’s mini-batch loop is plain Python, but each batch does only a handful of matrix multiplications on a tiny network, so the NumPy calls themselves dominate over the loop overhead; PyTorch’s per-batch loop involves autograd graph construction and teardown on every step, a fairly fixed per-batch cost that matters far more for a network this small than it would for a larger one where the actual computation would dominate instead. PyTorch’s flexibility, an explicit, inspectable training loop with autograd, comes with a real per-step overhead that a

network this size cannot amortize away, a trade-off that looks very different once real convolutions enter the picture in Sections 4 and 5.

Which framework hid the most, and which gave the most control: Keras hid essentially everything here too, the loss computation, the class-weighting mechanics, the backward pass, and the training loop are all one `compile()` call and one `fit()` call. PyTorch hid only the backward pass, keeping the training loop and the loss-weighting mechanism explicit, and that explicitness is exactly what surfaced the `pos_weight` asymmetry above, a detail that would have been invisible inside a packaged `fit()` call. Scratch hid nothing, at the direct cost of being the implementation with the most code to write and debug.

4 Fashion-MNIST: 10-Class Image Classification (CNN)

4.1 Problem description

This application classifies a 28×28 grayscale image of a clothing item into one of ten categories (T-shirt/top, trouser, pullover, dress, coat, sandal, shirt, sneaker, bag, ankle boot). Unlike the diabetes application, the input here has real 2-D spatial structure, neighboring pixels are related, and a shape like a sleeve or a shoe sole is a local pattern that recurs at different positions across the image, exactly the setting Lecture 04 gives for why a CNN, not an MLP, is the right architecture: local connectivity and weight sharing let a small set of learned kernels detect the same pattern anywhere in the image.

4.2 Dataset

Fashion-MNIST is loaded via `keras.datasets.fashion_mnist.load_data()`, which downloads and caches the dataset automatically on first use, no manual file management required. It ships as 60,000 training images and 10,000 test images, each 28×28 pixels, single channel, pixel values in $[0, 255]$ before normalization. Every one of the ten classes has exactly 6,000 training images and 1,000 test images (Figure 10), a deliberately balanced benchmark dataset, so no class-imbalance correction is needed here, unlike the diabetes application: a model that always predicted one class would score only 10% accuracy, not a deceptively high number the way it would on an imbalanced target.

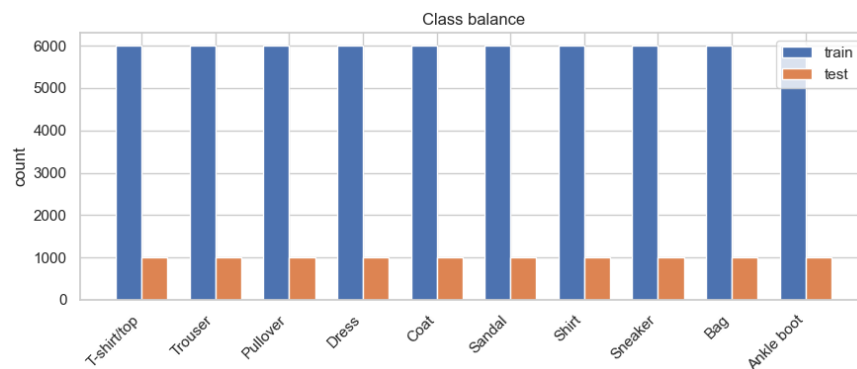


Figure 10: Fashion-MNIST class balance, train and test. Every class has exactly 6,000 training and 1,000 test images.

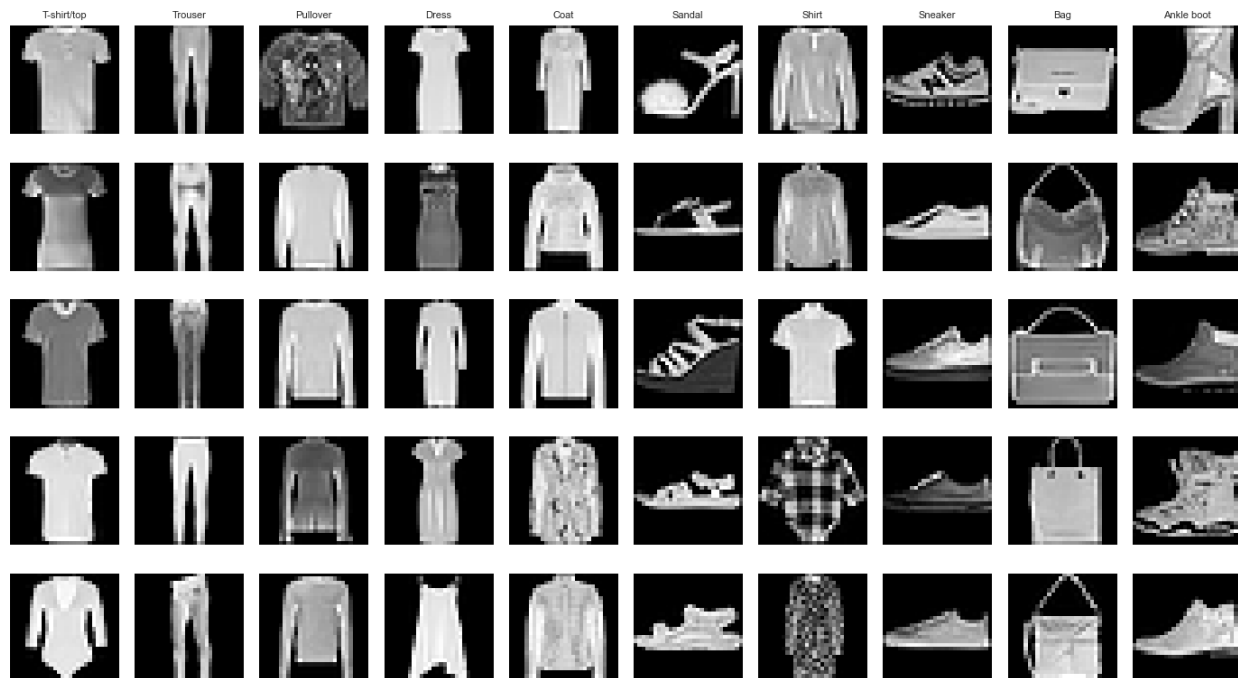


Figure 11: Five sample images per class. The classes are visually distinct in obvious cases and genuinely similar in others (shirt vs. pullover vs. coat), which is a reasonable source of the classification errors seen later.

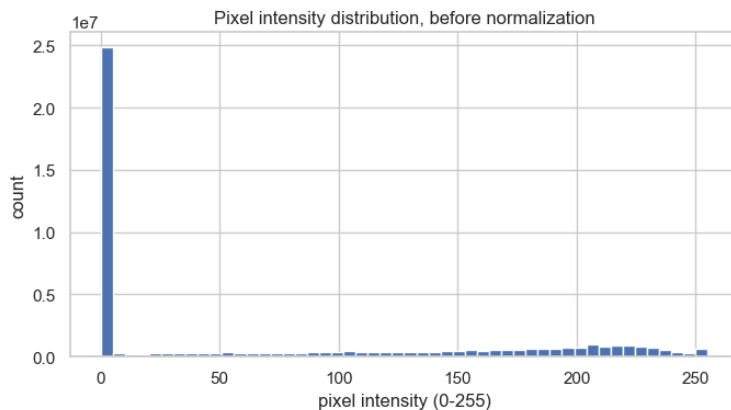


Figure 12: Pixel intensity distribution before normalization. The large spike at 0 reflects how much of every image is plain black background around a centered clothing item.

4.3 Representation

Pixel values are cast to `float32` and divided by 255, mapping the $[0, 255]$ integer range to $[0, 1]$. Two array layouts are kept side by side for the rest of the notebook: channel-first, $(N, 1, 28, 28)$, for scratch and PyTorch, and channel-last, $(N, 28, 28, 1)$, for Keras's default convolution layout. No further cleaning is needed, since this is a curated benchmark dataset with no missing values, no sentinel codes, and no schema inconsistencies to resolve.

4.4 Shared architecture

All three implementations build the same network:

$28 \times 28 \times 1 \rightarrow \text{Conv}(1 \rightarrow 32, k=3, \text{pad}=1) \rightarrow \text{ReLU} \rightarrow \text{MaxPool}(2) \rightarrow \text{Conv}(32 \rightarrow 64, k=3, \text{pad}=1) \rightarrow \text{ReLU} \rightarrow \text{MaxPool}(2)$
 $\rightarrow \text{Flatten}(64 \times 7 \times 7 = 3136) \rightarrow \text{Linear}(3136 \rightarrow 128) \rightarrow \text{ReLU} \rightarrow \text{Linear}(128 \rightarrow 10)$

with shape trace, for a batch of size B :

$(B, 1, 28, 28) \rightarrow (B, 32, 28, 28) \rightarrow (B, 32, 14, 14) \rightarrow (B, 64, 14, 14) \rightarrow (B, 64, 7, 7) \rightarrow (B, 3136) \rightarrow (B, 128) \rightarrow (B, 10)$

Batch size 256, 10 epochs, learning rate 0.05, and plain SGD are fixed once here and reused unchanged across all three implementations.

4.5 Three implementations

4.5.1 From scratch, NumPy only

Every operation, convolution, pooling, the dense layers, the loss, and every gradient, is implemented by hand. Convolution is the operation worth being concrete about: a 3×3 kernel is lined up against a 3×3 image patch, the nine overlapping (kernel value, pixel) pairs are multiplied and summed into one output value, and the kernel slides across every position to build the full feature map, the same kernel reused at every position, weight sharing, which is what lets one small kernel generalize across the whole image instead of only recognizing a pattern in the one spot it happened to be trained on. A direct position-by-position loop implementing that operation was too slow to be practical, the second convolutional layer alone took over two seconds per batch on this network's real layer sizes, so the notebook's `conv2d_forward/conv2d_backward` instead use an `im2col` approach: every patch the kernel will ever touch is extracted up front as a zero-copy strided view and stacked into one matrix, turning the whole convolution into a single matrix multiplication, the exact same computation as the naive loop, done for every position simultaneously rather than one at a time, about 40 times faster on this network's real layer sizes. A numerical gradient check confirms the hand-derived convolution and pooling backward passes against a finite-difference approximation before any full training run; this check caught a real bug during development, a redundant $1/N$ division in the dense-layer backward pass that had been silently shrinking every dense-layer gradient (convolution-layer gradients, computed by a different code path with no such division, were unaffected), fixed before the results reported below were produced. Mini-batch gradient descent (batch size 256, 235 batches per epoch) trains for 10 epochs.

4.5.2 TensorFlow/Keras

The identical architecture becomes an eight-layer `Sequential` definition (`Conv2D(32,3,padding="same",activation="relu")`, `MaxPooling2D(2)`, `Conv2D(64,...)`, `MaxPooling2D(2)`, `Flatten()`, `Dense(128,"relu")`, `Dense(10)`), compiled with `SGD(learning_rate=0.05)` and `SparseCategoricalCrossentropy(from_logits=True)`, trained with one `model.fit(X_train_hwc, y_train, batch_size=256, epochs=10)` call. `model.summary()` reports 421,642 parameters, matching scratch exactly. The convolution and pooling loops that scratch writes out explicitly have disappeared entirely from this implementation's own code, encapsulated instead inside the `Conv2D/MaxPooling2D` layer objects.

4.5.3 PyTorch

The same architecture becomes an `nn.Module` (FashionCNN) with a `self.features` block (`nn.Conv2d`, `nn.ReLU`, `nn.MaxPool2d`, twice) followed by a `self.classifier` block (`nn.Flatten`, `nn.Linear(3136,128)`, `nn.ReLU`, `nn.Linear(128,10)`), returning raw logits consumed by `nn.CrossEntropyLoss()`. The training loop stays explicit (`zero_grad` $\rightarrow$ forward $\rightarrow$ loss $\rightarrow$ backward() $\rightarrow$ `optimizer.step()`), with `torch.optim.SGD` at the same learning rate. The resulting parameter count, 421,642, matches both other implementations exactly, confirmed by a direct assertion in the notebook.

4.6 Component-by-implementation table

Table 5: Fashion-MNIST: component-by-implementation table

Component	Scratch	Keras	PyTorch
Data loading	Keras loader, shared NumPy arrays	same	same
Preprocessing	manual /255, channel-first reshape	/255, channel-last (default)	/255, channel-first
Model definition	explicit functions + a params dict	<code>Sequential([...])</code>	<code>nn.Module</code> subclass
Convolution	<code>conv2d_forward/backward</code> (<code>im2col</code>)	<code>Conv2D(...)</code>	<code>nn.Conv2d(...)</code>
Activation	<code>relu()</code>	<code>activation="relu"</code>	<code>nn.ReLU()</code>
Pooling	<code>max_pool2d_forward/backward</code>	<code>MaxPooling2D(2)</code>	<code>nn.MaxPool2d(2)</code>
Flatten	<code>flatten_forward/backward</code> (reshape)	<code>Flatten()</code>	<code>nn.Flatten()</code>
Dense/Linear layer	<code>linear_forward/backward</code>	<code>Dense(...)</code>	<code>nn.Linear(...)</code>
Loss	manual softmax cross-entropy	<code>SparseCategoricalCrossentropy</code> (<code>from_logits=True</code>)	<code>nn.CrossEntropyLoss()</code>
Gradient	manual backprop	<code>autodiff</code> (implicit in <code>fit</code>)	<code>loss.backward()</code> (autograd)
Optimizer	manual $W \leftarrow lr * dW$	<code>optimizer="sgd"</code>	<code>torch.optim.SGD(...)</code>
Training loop	explicit for epoch / for batch	<code>model.fit(...)</code>	explicit for epoch / for batch
Evaluation	manual metric functions	<code>model.evaluate</code> -equivalent (manual)	manual (<code>model.eval()</code> , no-grad)
Prediction	<code>forward(X_test, params)</code>	<code>model.predict(...)</code>	<code>model(x_test)</code> under <code>torch.no_grad()</code>

4.7 Training curves and evaluation

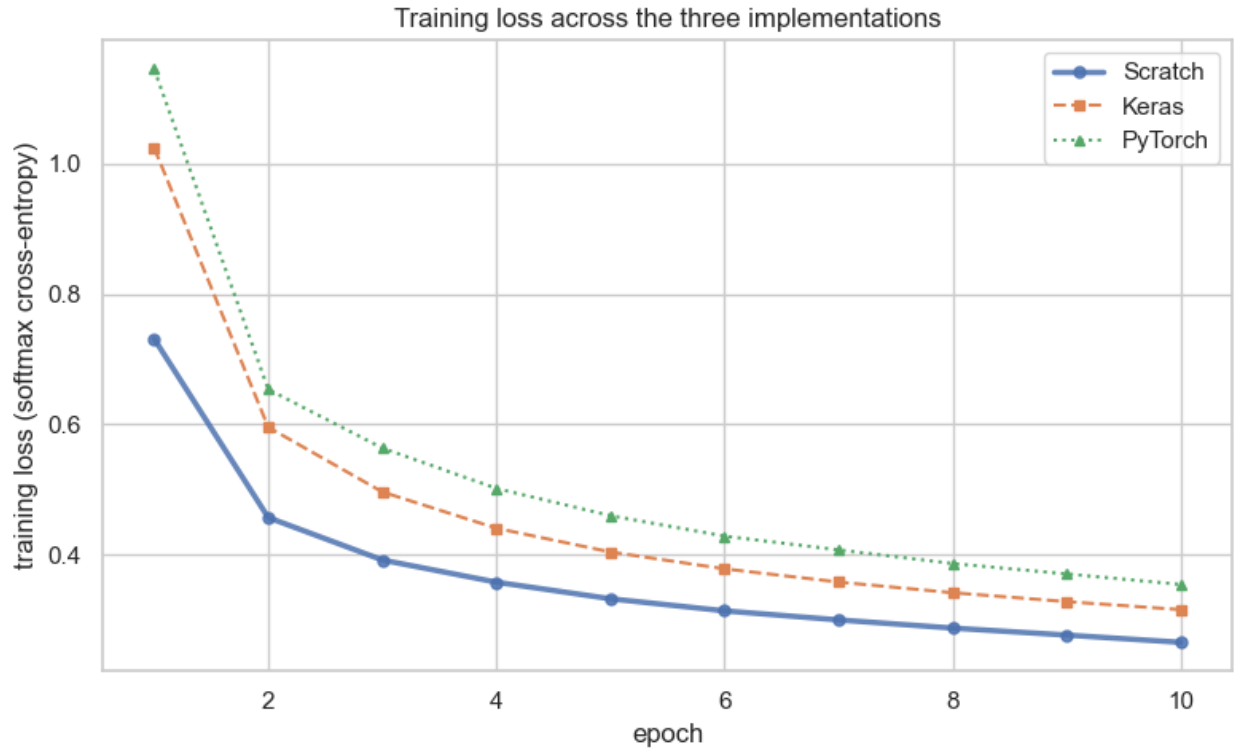


Figure 13: Training loss across the three implementations.

Scratch											
Actual	T-shirt/top	830	2	19	21	8	1	109	0	10	0
	Trouser	0	984	1	11	2	0	1	0	1	0
	Pullover	10	3	837	10	86	0	53	0	1	0
	Dress	12	12	8	894	44	0	26	0	4	0
	Coat	1	1	69	15	866	0	48	0	0	0
	Sandal	0	0	0	0	0	969	0	20	1	10
	Shirt	99	3	78	25	96	0	692	0	7	0
	Sneaker	0	0	0	0	0	10	0	974	1	15
	Bag	1	1	1	4	4	2	4	5	978	0
	Ankle boot	0	0	0	0	0	4	1	46	0	949
Keras											
Actual	T-shirt/top	849	3	35	36	5	0	63	0	9	0
	Trouser	2	980	1	12	2	0	1	0	2	0
	Pullover	15	2	920	14	39	0	8	0	2	0
	Dress	12	26	13	902	24	0	19	0	4	0
	Coat	1	2	249	40	888	0	19	0	1	0
	Sandal	0	0	0	1	0	924	0	58	1	16
	Shirt	143	4	246	32	99	0	462	0	14	0
	Sneaker	0	0	0	0	0	4	0	983	1	12
	Bag	3	1	15	6	1	1	3	6	964	0
	Ankle boot	0	0	1	0	0	3	0	65	0	931
PyTorch											
Actual	T-shirt/top	826	1	19	26	8	3	93	0	24	0
	Trouser	3	971	0	16	5	1	2	0	2	0
	Pullover	17	2	733	12	170	0	51	0	15	0
	Dress	29	12	5	853	65	0	29	0	7	0
	Coat	1	1	56	16	884	0	36	0	6	0
	Sandal	1	0	0	1	0	960	0	28	1	9
	Shirt	148	1	87	25	149	1	562	0	27	0
	Sneaker	0	0	0	0	0	22	0	954	1	23
	Bag	0	1	2	2	5	1	2	5	982	0
	Ankle boot	0	0	0	0	0	12	0	52	1	935

Figure 14: Confusion matrices for the three implementations on the Fashion-MNIST test set (10,000 images). The shirt/pullover/coat cluster is the most common source of confusion in all three, consistent with how visually similar those classes are (Figure 10's neighboring figure).

Table 6: Fashion-MNIST: quantitative comparison, test set

Implementation	Params	Train time (s)	Final loss	Accuracy	Precision (macro)	Recall (macro)
Scratch	421,642	1,330.5	0.2655	0.8973	0.8976	0.8973
Keras	421,642	75.6	0.3157	0.8603	0.8695	0.8603
PyTorch	421,642	191.5	0.3544	0.8660	0.8676	0.8660

Macro F1 for the same three implementations, in the same order: 0.8970 (scratch), 0.8565 (Keras), 0.8641 (PyTorch).

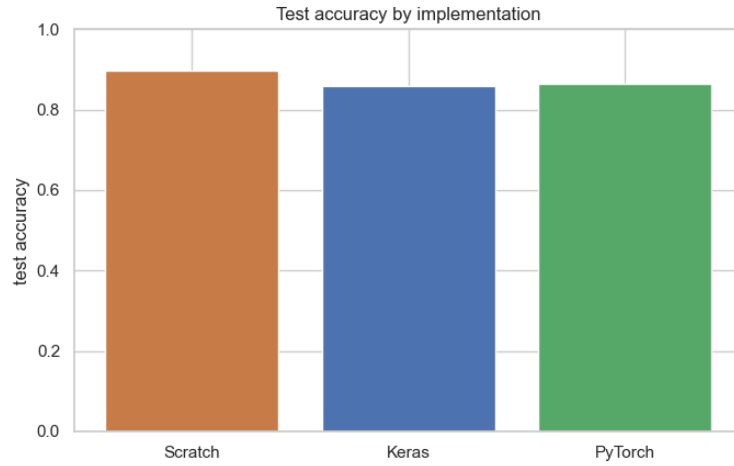


Figure 15: Test accuracy by implementation. Scratch wins here, the only application in this report where it does.

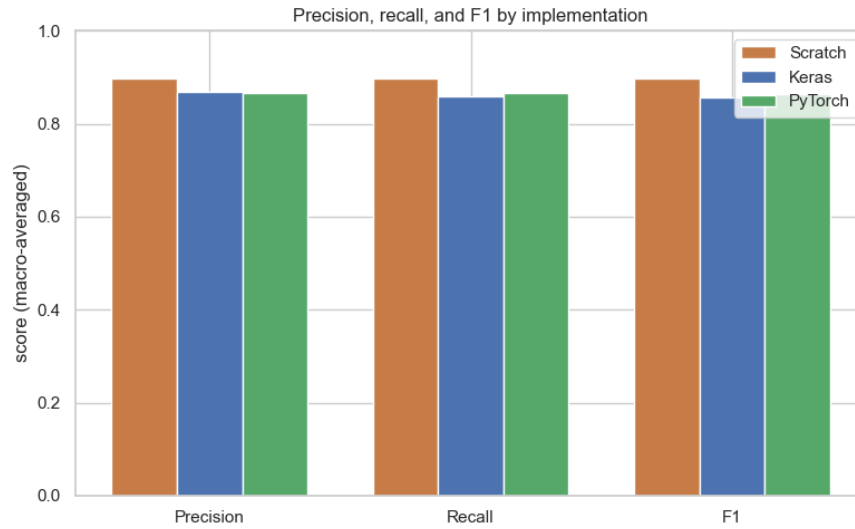


Figure 16: Precision, recall, and F1 (macro-averaged) by implementation.

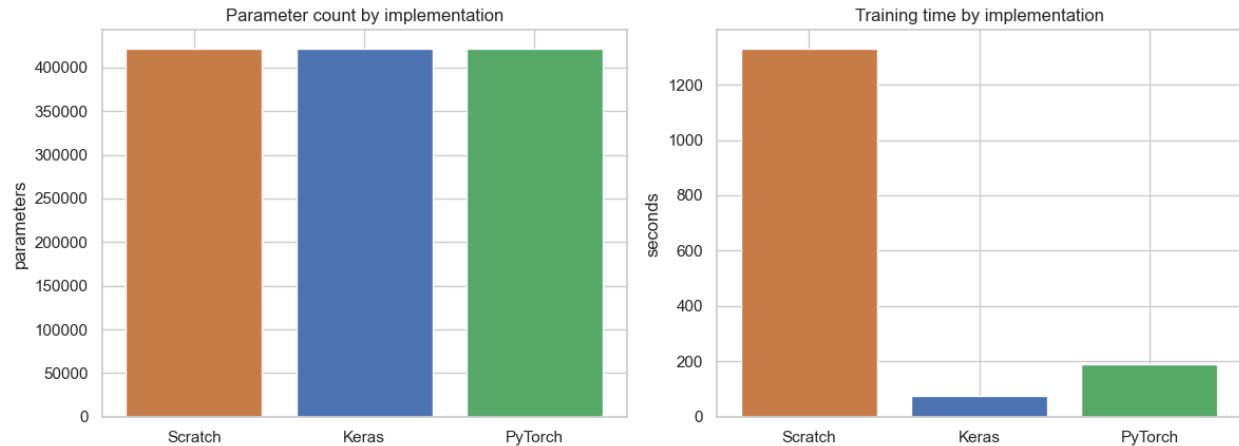


Figure 17: Parameter count (identical across all three) and training time by implementation. Scratch’s training time is on a dramatically different scale from the other two.

4.8 Comparison discussion

By test accuracy, scratch actually comes out ahead: 0.897 against PyTorch’s 0.866 and Keras’s 0.860. That is a genuinely interesting result, since scratch is not usually the implementation expected to win. The most likely explanation is initialization and randomness, not anything about the implementation being fundamentally better: scratch uses explicit He initialization with a fixed seed, while Keras and PyTorch each apply their own default initializer, also ReLU-appropriate but not guaranteed to produce the same starting point. Ten epochs is not a lot of training, so a difference in where each implementation starts can still show up meaningfully in where it ends up. All three are within about four points of each other on accuracy, small enough to be consistent with three implementations of the same architecture rather than three genuinely different models, echoing the diabetes application’s own spread.

Training time tells the more dramatic story here. Scratch took 1,330.5 seconds (about 22 minutes), Keras took 75.6 seconds, and PyTorch took 191.5 seconds, scratch roughly 17.6 times slower than Keras and 6.9 times slower than PyTorch. That gap makes sense given what each implementation is actually doing under the hood: Keras’s `fit()` runs the entire training loop inside a compiled graph, with convolution implemented by heavily optimized, vectorized routines and essentially no per-batch Python overhead. PyTorch’s explicit training loop carries some per-batch Python and autograd overhead, but its convolution operation is still a compiled, optimized routine. Scratch’s `conv2d_forward/conv2d_backward`, even with the `im2col` speedup over a naive position-by-position loop, are still plain NumPy matrix multiplies issued from a Python training loop, with none of the operator fusion or lower-level optimization a mature deep learning framework applies to convolution specifically. The `im2col` rewrite made training feasible at all; it does not close the gap with frameworks built and optimized specifically around this operation. This is a sharp contrast with the diabetes MLP in Section 3.9, where scratch was comparably fast to the frameworks and PyTorch was in fact the slowest implementation: once a network’s forward pass genuinely involves convolution rather than only small dense-layer matrix multiplies, the speed picture flips entirely.

Which framework hid the most, and which gave the most control: Keras hid essentially everything, the convolution, the pooling, the backward pass, and the training loop are all one `Sequential` definition and one `fit()` call. PyTorch hid the backward pass via autograd but kept the training loop and the layer composition explicit. Scratch hid nothing, every multiply, every gradient, and every parameter update is code that was written and can be read line by line. The cost of that visibility, in this application, is a very real difference

in wall-clock training time, exactly the trade-off the three abstraction levels are meant to represent: more convenience and speed at the cost of visibility, or more visibility and control at the cost of convenience and speed.

5 CIFAR-10: 10-Class Image Classification (CNN)

5.1 Problem description

This application classifies a 32×32 RGB color photograph into one of ten object categories (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck). It uses the same architecture family as Fashion-MNIST, a CNN, for the same reason, real 2-D spatial structure that a convolution can exploit, but with three input channels instead of one and a task that is substantially harder: distinguishing a cat from a dog, or an automobile from a truck, in a real photograph compressed to 32×32 pixels, is a genuinely harder visual problem than distinguishing clothing silhouettes on a plain background.

5.2 Dataset

CIFAR-10 is loaded via `keras.datasets.cifar10.load_data()`, which downloads and caches the dataset automatically on first use (~ 170 MB, a one-time download). It ships as 50,000 training images and 10,000 test images, each 32×32 pixels, three channels, pixel values in $[0, 255]$ before normalization. Every one of the ten classes has exactly 5,000 training images and 1,000 test images, another deliberately balanced benchmark dataset requiring no class-weighting correction.

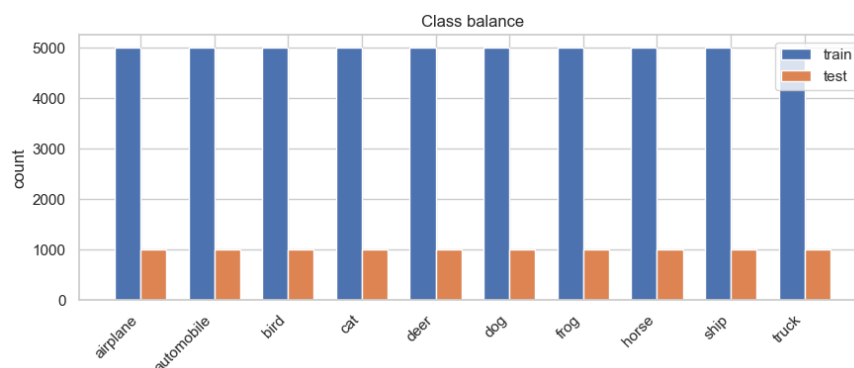


Figure 18: CIFAR-10 class balance, train and test. Every class has exactly 5,000 training and 1,000 test images.

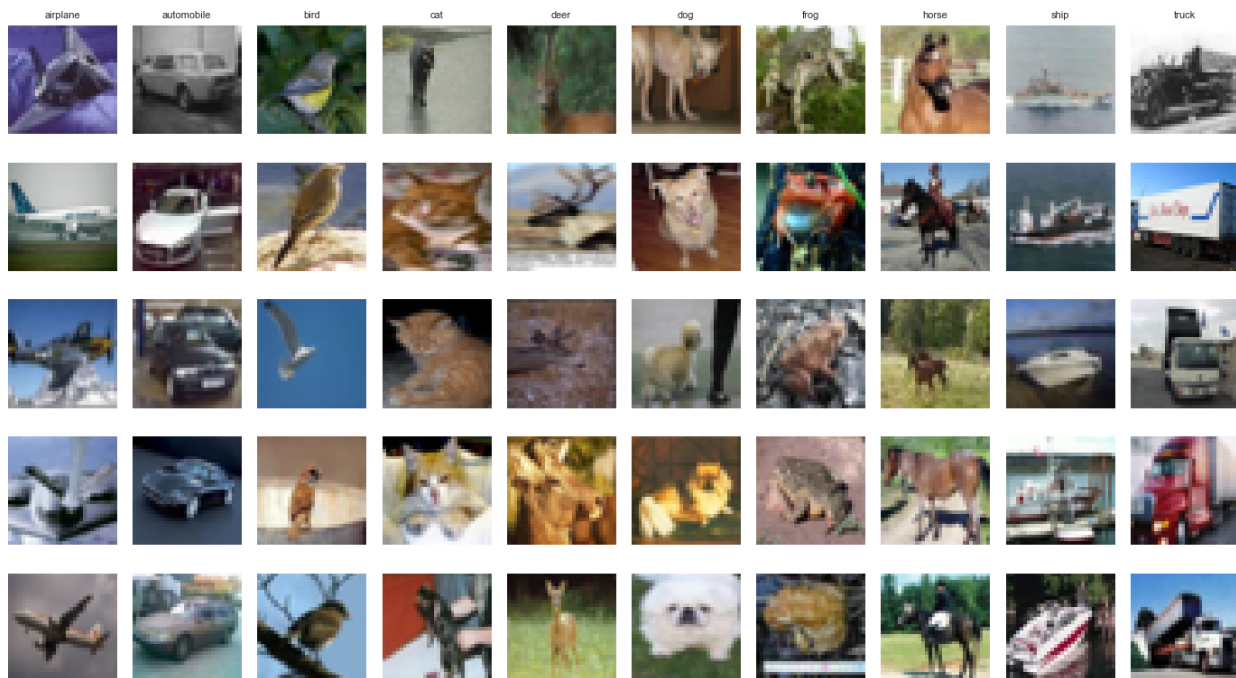


Figure 19: Five sample images per class. These are real color photographs at only 32×32 resolution, small enough that some category pairs (automobile/truck, cat/dog) are genuinely ambiguous even to a human glance.

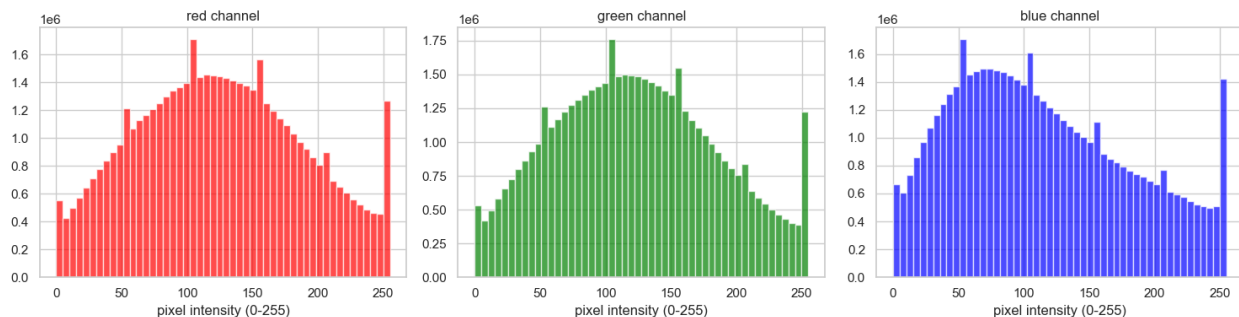


Figure 20: Per-channel (red, green, blue) pixel intensity distribution before normalization. All three channels span the full 0-255 range with a broadly similar shape, unlike Fashion-MNIST’s single-channel histogram, which had a dominant spike at 0 from plain black background.

5.3 Representation

Pixel values are cast to `float32` and divided by 255. CIFAR-10 ships channel-last, $(N, 32, 32, 3)$, matching Keras’s default layout directly with no transpose needed; scratch and PyTorch instead use a transposed channel-first copy, $(N, 3, 32, 32)$. No further cleaning is needed, the same curated-benchmark reasoning as Fashion-MNIST.

5.4 Shared architecture

All three implementations build the same network, a direct instance of the CNN architecture the course tutorial material works out in full for a $3 \times 32 \times 32$ input:

$3 \times 32 \times 32 \rightarrow \text{Conv}(3 \rightarrow 32, k=3, \text{pad}=1) \rightarrow \text{ReLU} \rightarrow \text{MaxPool}(2) \rightarrow \text{Conv}(32 \rightarrow 64, k=3, \text{pad}=1) \rightarrow \text{ReLU} \rightarrow \text{MaxPool}(2)$
 $\rightarrow \text{Flatten}(64 \times 8 \times 8 = 4096) \rightarrow \text{Linear}(4096 \rightarrow 128) \rightarrow \text{ReLU} \rightarrow \text{Linear}(128 \rightarrow 10)$

with shape trace, for a batch of size B :

$(B, 3, 32, 32) \rightarrow (B, 32, 32, 32) \rightarrow (B, 32, 16, 16) \rightarrow (B, 64, 16, 16) \rightarrow (B, 64, 8, 8) \rightarrow (B, 4096) \rightarrow (B, 128) \rightarrow (B, 10)$

matching the tutorial's own worked shape progression exactly. Batch size 256, 12 epochs, learning rate 0.05, and plain SGD are fixed once here and reused unchanged across all three implementations.

5.5 Three implementations

5.5.1 From scratch, NumPy only

The scratch implementation reuses the same `conv2d_forward/backward (im2col)`, `max_pool2d_forward/backward`, `relu()`, and `softmax-cross-entropy` machinery built for Fashion-MNIST, made explicitly 3-channel-aware: the first kernel's shape is $(32, 3, 3, 3)$ rather than $(32, 1, 3, 3)$, since the input now has three input channels instead of one. A numerical gradient check on a synthetic 3-input-channel sample confirms the convolution backward pass is correct specifically for the multi-channel case, not only the single-channel case Fashion-MNIST exercised, before any full training run. The same redundant $1/N$ dense-layer division bug fixed for Fashion-MNIST was fixed here too, before this notebook was ever executed, so no training run here was affected by it. Mini-batch gradient descent (batch size 256, 196 batches per epoch) trains for 12 epochs.

5.5.2 TensorFlow/Keras

The identical architecture becomes the same eight-layer `Sequential` shape as Fashion-MNIST, with `Conv2D(32, 3, ...)` taking a $(32, 32, 3)$ input directly, no channel-order conversion needed since CIFAR-10 already ships in Keras's preferred layout. Compiled with `SGD(learning_rate=0.05)` and `SparseCategoricalCrossentropy(from_logits=True)`, trained with `model.fit(X_train_hwc, y_train, batch_size=256, epochs=12)` call. `model.summary()` reports 545,098 parameters, matching scratch exactly.

5.5.3 PyTorch

The same architecture becomes an `nn.Module (Cifar10CNN)`, structurally identical to Fashion-MNIST's `FashionCNN` except for the first convolution's input channel count (3 instead of 1) and the flattened feature size ($64 \times 8 \times 8 = 4096$ instead of $64 \times 7 \times 7 = 3136$, since CIFAR-10's larger 32×32 input survives two stride-2 poolings at a larger spatial size than Fashion-MNIST's 28×28 does). The explicit training loop and `nn.CrossEntropyLoss()` carry over unchanged from Fashion-MNIST. The resulting parameter count, 545,098, matches both other implementations exactly.

5.6 Component-by-implementation table

Table 7: CIFAR-10: component-by-implementation table

Component	Scratch	Keras	PyTorch
Data loading	Keras loader, shared NumPy arrays	same	same
Preprocessing	manual /255, channel-first reshape	/255, channel-last (as shipped, no transpose)	/255, channel-first (transposed)
Model definition	explicit functions + a params dict	<code>Sequential([...])</code>	<code>nn.Module</code> subclass
Convolution	<code>conv2d_forward/backward</code> (im2col, 3-channel-aware)	<code>Conv2D(...)</code>	<code>nn.Conv2d(...)</code>
Activation	<code>relu()</code>	<code>activation="relu"</code>	<code>nn.ReLU()</code>
Pooling	<code>max_pool2d_forward/backward</code>	<code>MaxPooling2D(2)</code>	<code>nn.MaxPool2d(2)</code>
Flatten	<code>flatten_forward/backward</code> (reshape)	<code>Flatten()</code>	<code>nn.Flatten()</code>
Dense/Linear layer	<code>linear_forward/backward</code>	<code>Dense(...)</code>	<code>nn.Linear(...)</code>
Loss	manual softmax cross-entropy	<code>SparseCategoricalCrossentropy</code> (<code>from_logits=True</code>)	<code>nn.CrossEntropyLoss()</code>
Gradient	manual backprop	<code>autodiff</code> (implicit in <code>fit</code>)	<code>loss.backward()</code> (autograd)
Optimizer	manual $W \leftarrow lr * dW$	<code>optimizer="sgd"</code>	<code>torch.optim.SGD(...)</code>
Training loop	explicit for epoch / for batch	<code>model.fit(...)</code>	explicit for epoch / for batch
Evaluation	manual metric functions	<code>model.evaluate</code> -equivalent (manual)	manual (<code>model.eval()</code> , no-grad)
Prediction	<code>forward(X_test, params)</code>	<code>model.predict(...)</code>	<code>model(x_test)</code> under <code>torch.no_grad()</code>

5.7 Training curves and evaluation

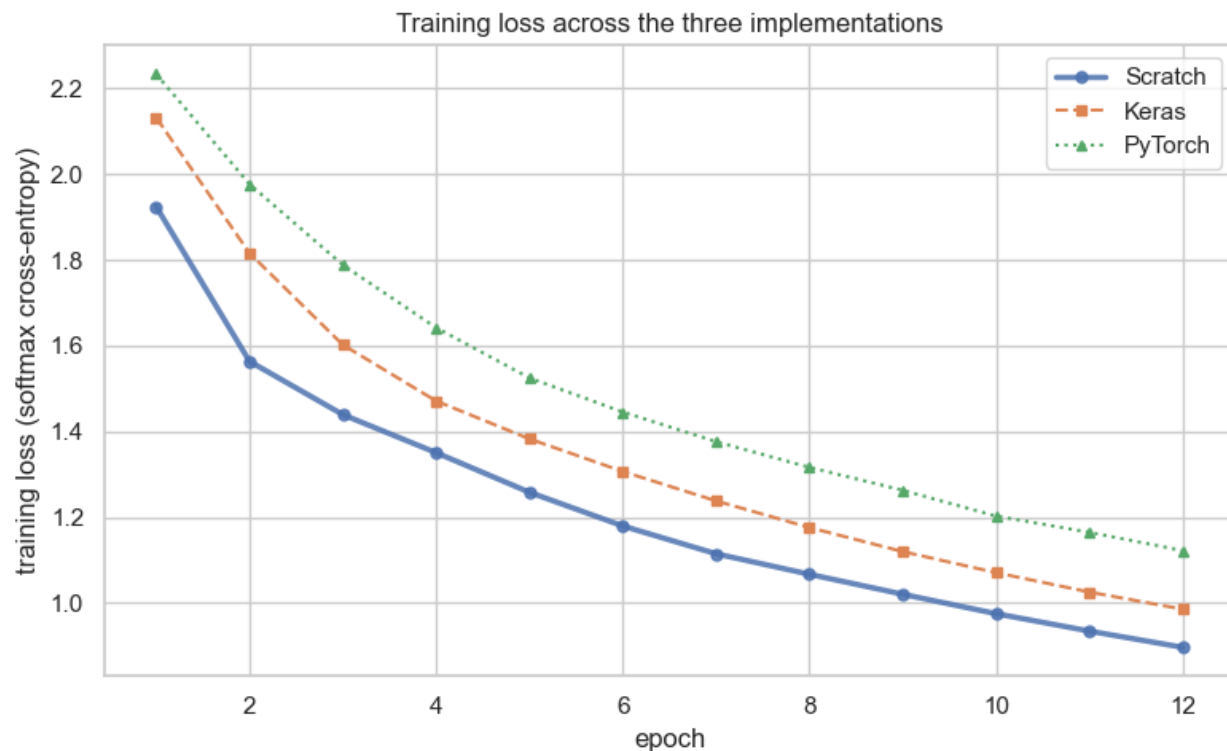


Figure 21: Training loss across the three implementations.

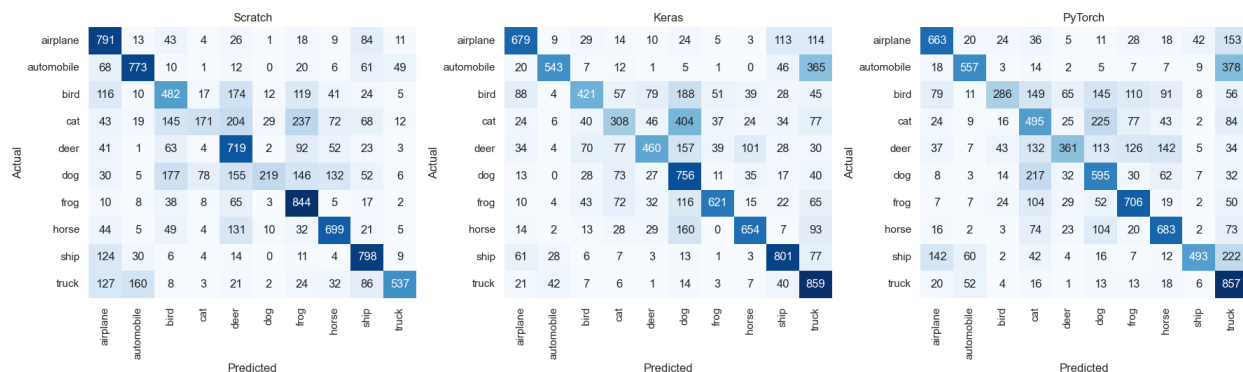


Figure 22: Confusion matrices for the three implementations on the CIFAR-10 test set (10,000 images). Cat/dog and automobile/truck are the most commonly confused pairs in all three, consistent with the visual ambiguity noted in the sample grid above.

Table 8: CIFAR-10: quantitative comparison, test set

Implementation	Params	Train time (s)	Final loss	Accuracy	Precision (macro)	Recall (macro)
Scratch	545,098	1,875.8	0.8965	0.6033	0.6335	0.6033
Keras	545,098	103.3	0.9854	0.6102	0.6477	0.6102
PyTorch	545,098	244.1	1.1220	0.5696	0.6163	0.5696

Macro F1 for the same three implementations, in the same order: 0.5794 (scratch), 0.6078 (Keras), 0.5655 (PyTorch).

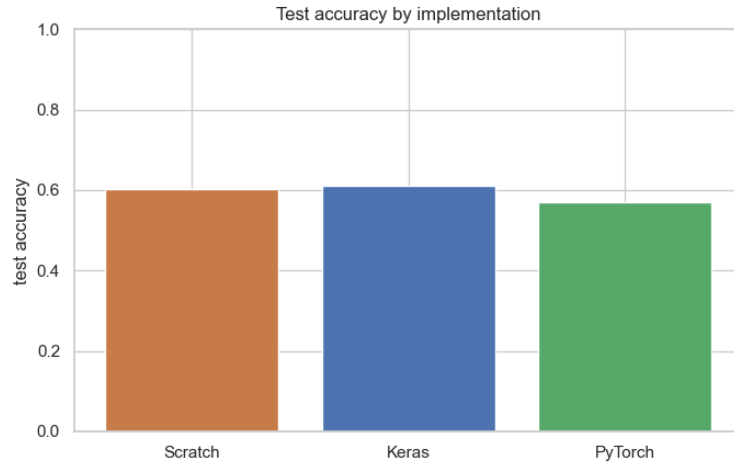


Figure 23: Test accuracy by implementation. Absolute accuracy is much lower than Fashion-MNIST's, the expected effect of a genuinely harder task.

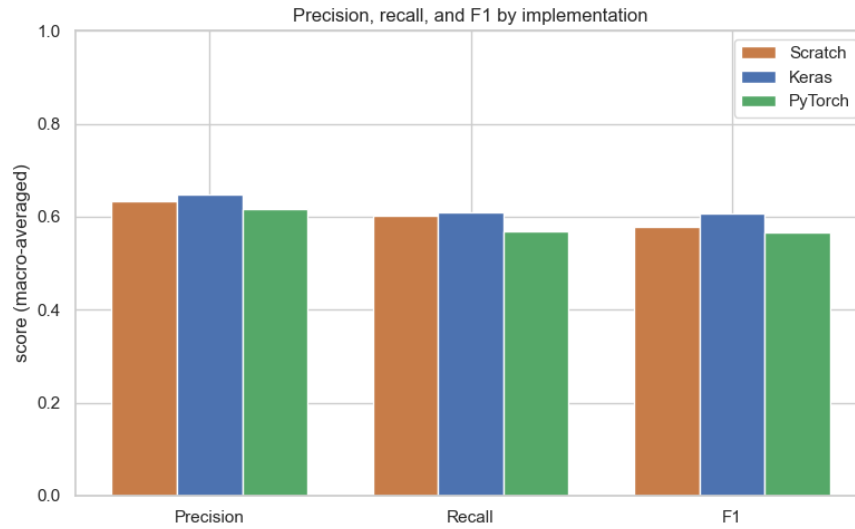


Figure 24: Precision, recall, and F1 (macro-averaged) by implementation.

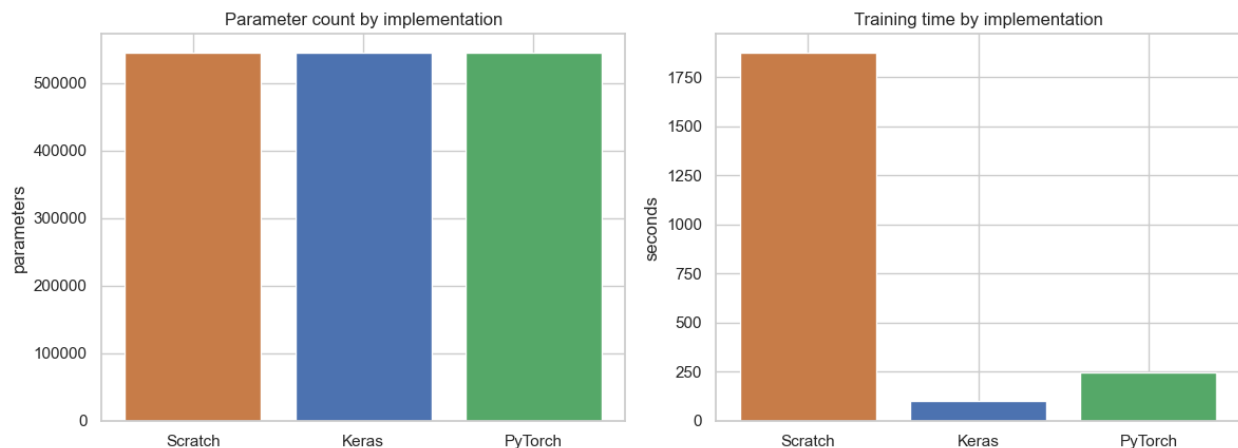


Figure 25: Parameter count (identical across all three) and training time by implementation. The scratch/framework training-time gap is the widest of the three applications in this report.

5.8 Comparison discussion

Keras comes out slightly ahead on both accuracy (0.610) and F1 (0.608), with scratch close behind (0.603 accuracy, 0.579 F1) and PyTorch trailing a little further (0.570 accuracy, 0.566 F1). All three sit within about four points of each other, the same order of spread seen for both other applications in this report, consistent with three implementations of the same architecture rather than three different models. The absolute accuracy here, in the low 60s, is much lower than Fashion-MNIST’s high 80s with the same architecture, and that is expected: these are real color photographs compressed to 32×32 pixels, category pairs like cat/dog and automobile/truck are genuinely hard to separate even at that resolution, and a two-convolution-layer network with no data augmentation and only 12 epochs of plain SGD is a modest model for a task at this level of difficulty. Accuracy in the low 60s here is not a sign anything is broken; it is roughly what this architecture is capable of on this data without the training tricks, augmentation, more epochs, a bigger network, a fancier optimizer, deliberately left out to keep the three implementations easy to compare against each other.

Training time repeats the pattern first seen in Fashion-MNIST, scratch is dramatically slower than the other two, and the gap widens further here. Scratch took 1,875.8 seconds (about 31 minutes) to train. Keras took 103.3 seconds. PyTorch took 244.1 seconds. Scratch was roughly 18.2 times slower than Keras and 7.7 times slower than PyTorch, a wider multiple than Fashion-MNIST’s 17.6x/6.9x, despite this network having only modestly more parameters (545,098 against 421,642). That points at input size and channel count, three color channels instead of one, and a 32×32 image instead of 28×28 , rather than parameter count, as the real driver of scratch’s `im2col` convolution cost: both changes increase the amount of work the first convolution’s matrix multiply has to do, and scratch pays that cost directly in wall-clock time with no framework-level optimization to absorb it, while Keras’s and PyTorch’s compiled convolution operations scale far more gracefully. Read together with Section 4.8, a clean three-point progression emerges across this report’s three applications: the diabetes MLP (Section 3.9), scratch comparable in speed to the frameworks, PyTorch actually slowest; the grayscale CNN, scratch roughly 17.6x slower than Keras; the RGB CNN, scratch roughly 18.2x slower than Keras. Training-time cost from building a network by hand grows specifically with how much real convolution work the architecture asks for, not with parameter count alone.

Which framework hid the most, and which gave the most control, is the same answer as Fashion-MNIST’s, for the same reasons: Keras hid the convolution, the pooling, the backward pass, and the training loop behind one `Sequential` definition and one `fit()` call; PyTorch hid only the backward pass; scratch hid

nothing. The three-application pattern across this report is now clear: accuracy differences between scratch, Keras, and PyTorch stay small, within a few points, whenever the architecture is genuinely held fixed, the expected result and the main point of building the same network three times. Training time differences, on the other hand, can be large and depend heavily on what the architecture actually asks the hardware to do, negligible for a tiny fully-connected network, large and specifically tied to convolution workload for a CNN, a from-scratch NumPy implementation, however correct, cannot match a mature framework's operator-level optimization for an operation like convolution without adopting the same kind of low-level optimization work those frameworks represent.

6 Cross-Application Comparison

Table 9 summarizes all three applications by test accuracy, the primary metric used throughout this report since none of the three tasks is regression, across all three implementations. Every cell is a transposition of the per-application results tables in Sections 3, 4, and 5, not new analysis.

Table 9: Cross-application summary: test accuracy, all 3 applications x 3 implementations

Application	Implementation	Parameters	Accuracy
Diabetes	Scratch	2,177	0.7310
	Keras	2,177	0.7533
	PyTorch	2,177	0.7819
Fashion-MNIST	Scratch	421,642	0.8973
	Keras	421,642	0.8603
	PyTorch	421,642	0.8660
CIFAR-10	Scratch	545,098	0.6033
	Keras	545,098	0.6102
	PyTorch	545,098	0.5696

Bold marks the winning implementation in each application. Parameter counts are identical across all three implementations within an application, confirmed by a direct assertion in every notebook.

Table 10: Cross-application training time, seconds, all 3 applications x 3 implementations

Application	Scratch	Keras	PyTorch
Diabetes	69.4	24.7	311.0
Fashion-MNIST	1,330.5	75.6	191.5
CIFAR-10	1,875.8	103.3	244.1

Bold marks the fastest implementation in each application. Keras is fastest in every application. Scratch is fastest of the two non-Keras implementations only on diabetes, its tiny MLP; scratch is the slowest implementation by a wide margin on both CNN applications.

No single implementation wins every application, and the implementation that wins by the widest margin, PyTorch on diabetes, is also the slowest implementation on that same application, a direct illustration that accuracy and training speed are independent axes here, not two views of the same underlying quality.

7 Discussion

7.1 Did “same architecture, different abstraction level” actually hold?

Yes, on the axis it makes the strongest claim about. Every application’s parameter count matched exactly across all three implementations, verified by a direct assertion in each notebook rather than assumed, and every application’s accuracy spread across the three implementations stayed within about four to five points, the kind of gap explainable by initialization differences and the ordinary noise of training a fixed architecture

rather than evidence that one implementation is a fundamentally better learner. Table 9’s flipping winner, PyTorch on diabetes, scratch on Fashion-MNIST, Keras on CIFAR-10, is the pattern this claim predicts: if a particular framework’s implementation of “the same architecture” were secretly a stronger model, it would win consistently, not take turns.

7.2 Where the abstraction levels genuinely diverged

Training time is where the three abstraction levels stopped behaving like interchangeable views of the same computation. On the diabetes MLP, a tiny, fully dense network, scratch and Keras trained in a comparable amount of time, and PyTorch was actually the slowest implementation, its per-batch autograd construction overhead a fixed cost too large for a 2,177-parameter network’s real computation to amortize away. Once real convolutions entered the picture, Fashion-MNIST and CIFAR-10, that pattern reversed completely: scratch became the slowest implementation by a wide and widening margin (17.6x Keras on Fashion-MNIST, 18.2x Keras on CIFAR-10), because a from-scratch `im2col` convolution, however correctly implemented and however much faster than the naive position-loop it replaced, is still plain NumPy issued from a Python training loop, with none of a mature framework’s operator-level optimization for convolution specifically. The gap widened again between the two CNN applications in step with input size and channel count (CIFAR-10’s $32 \times 32 \times 3$ against Fashion-MNIST’s $28 \times 28 \times 1$), not with parameter count, which grew only modestly between the two (545,098 against 421,642). Visibility, correctness, and speed are three separate properties of an implementation, and this project’s own measured numbers show them trading off in exactly the direction Lecture 04 predicts, more of the first two costing real wall-clock time.

7.3 What building every training loop by hand exposed

Two findings surfaced specifically because this assignment kept scratch’s implementation fully visible rather than trusting a packaged `fit()` call, and both are worth naming because a framework absorbs them silently. First, the diabetes application’s PyTorch loss value (Section 3.9) sits on a genuinely different numeric scale from scratch’s and Keras’s, not because PyTorch trained a worse model, but because `BCEWithLogitsLoss(pos_weight=...)` reweights only the positive class’s loss term while scratch’s manual weighting and Keras’s `class_weight` both reweight every row. Three frameworks offering what looks like the same named feature, balanced class weighting, actually parameterize it differently underneath, exactly Lecture 04’s own warning that different names do not imply different concepts, read in reverse: sometimes the same name does hide a real mechanical difference, and the only way to know is to check the code, not trust the name. Second, the Fashion-MNIST scratch implementation’s own gradient-check cell caught a real bug during development, a redundant $1/N$ division in the dense-layer backward pass silently shrinking every dense-layer gradient, invisible from the outside since the network still trained and still produced a plausible-looking loss curve; only comparing the analytic gradient against a numerical finite-difference approximation, layer by layer, surfaced it. Both findings are the direct payoff of building the optimizer, the loss, and the gradient computation by hand rather than importing them: a packaged implementation would have absorbed either issue without ever surfacing it as something to explain.

8 Conclusion

This report built three intelligent systems, diabetes risk screening (MLP), Fashion-MNIST clothing recognition (CNN), and CIFAR-10 object recognition (CNN), each with a single architecture implemented three separate times, from scratch in NumPy with every forward pass, loss, backward pass, and parameter update derived and coded by hand, in TensorFlow/Keras, and in PyTorch. Every application’s parameter count matched exactly across all three implementations, and every application’s accuracy spread stayed within a

few points, confirming Lecture 04’s central claim that the conceptual architecture is unchanged across abstraction levels, only its expression changes. What genuinely differed, and differed by a large, measurable margin once real convolutions were involved, was training time: a from-scratch NumPy implementation, however correct, cannot match a mature framework’s operator-level optimization for an operation like convolution, a real cost of full visibility that a tiny MLP’s forward pass does not expose but a CNN’s does, and the gap widens further with input size and channel count rather than with parameter count. Keeping every implementation’s own training loop and loss computation fully visible, rather than trusting a packaged `fit()` call, is also what surfaced two concrete findings a framework would have absorbed silently: PyTorch’s `pos_weight` class-weighting mechanism scales the training loss differently from scratch’s and Keras’s symmetric weighting despite sharing the same balanced-weighting intent, and a redundant division bug in a from-scratch backward pass, caught only by a numerical gradient check comparing hand-derived gradients against a finite-difference approximation. Both are the kind of detail visible only to someone who wrote every gradient by hand, exactly the payoff building a CNN from scratch is meant to provide.

9 Reproducibility

9.1 Environment

All three notebooks were built and executed in a dedicated virtual environment at `assignment_04/.venv` (Python 3.10.11, registered as Jupyter kernel `assignment_04`), with the following key package versions: `numpy`, `pandas`, `scikit-learn` (metrics only), `matplotlib`, `seaborn`, `jupyter`, `nbconvert`, `nbformat`, `nbclient`, `tensorflow` 2.21.0, `torch` 2.14.0+cpu, `torchvision` 0.29.0+cpu. CPU-only PyTorch/torchvision wheels were installed deliberately, so the three-implementation training-time comparison is not confounded by one implementation running on a different device class than the others. `RANDOM_SEED = 42` is fixed in every notebook for every split, shuffle, and weight initialization (`np.random.seed`, Python’s own `random.seed`, `tf.random.set_seed`, and `torch.manual_seed`, all set together).

9.2 Re-running a notebook end to end

Each application’s notebook is self-contained and can be re-executed from its own directory with:

```
../.venv/Scripts/python.exe -m jupyter nbconvert \  
  --to notebook --execute --inplace \  
  notebook/<app>.ipynb --ExecutePreprocessor.timeout=1800
```

replacing `<app>` with `diabetes`, `fashion_mnist`, or `cifar10`. Fashion-MNIST and CIFAR-10 download their datasets automatically via `keras.datasets` on first run, no manual Kaggle step required, unlike the diabetes application’s CSV files, which must already be present in `diabetes/data/`. Each notebook ends by saving all three trained artifacts to its own `model/` directory:

- `scratch_weights.npz`: the from-scratch implementation’s trained parameters.
- `keras_model.keras`: the trained Keras model.
- `pytorch_model.pt`: the trained PyTorch model’s state dict.
- `feature_names.joblib` (diabetes only): the post-encoding feature column order.
- `input_schema.json`: documented input fields and target.

followed by a reload-and-verify cell that reloads all three artifacts and confirms their predictions match the in-memory versions computed during that same run. No API, web, or mobile layer is included in this assignment, so no additional run commands beyond notebook execution are required.